

Raagav Ramakrishnan

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ML Systems Engineering • AI for Healthcare • Medical Imaging

EDUCATION

International Institute of Information Technology, Hyderabad (IIIT-H) 2022 – 2026
B.Tech Honours in Electronics and Communication

The Millennium School, Dubai June 2022
High School Diploma CBSE — Grade 10: 93%, Grade 12: 95%

WORK EXPERIENCE

ML Research Engineer Dec. 2024 – Present
Cloud Ambassadors (Startup within the Google Ecosystem)

- Building a Glioma Detection and Segmentation pipeline for clinical workflows on GCP.
- Fine-tuned ProtBERT and ESM-1b models for mutation effect prediction; integrated Graph Neural Networks (Graphormer/MPNN) using protein contact maps to model structural dependencies.
- Developed and deployed a production-grade Dynamic Pricing Engine using FastAPI and containerized services. Built end-to-end pipelines on GCP using Kubeflow, BigQuery, and XGBoost.
- Implemented continuous training and automated inference workflows with Vertex AI, Pub/Sub triggers, and Cloud Build CI/CD, enabling seamless model versioning and rollout.

Undergraduate Researcher Jan. 2025 – Present
Data Sciences and Analytics Center (DSAC), IIIT Hyderabad *Advisor: Dr. Nimmi Rangaswamy*

- Studying how young adults *bricolage* fitness tools and social media practices to form identity, sustain motivation, and protect their well-being in gym cultures.
- Conducting field visits to local gyms and training spaces, supported by qualitative interviews and situated conversations with practitioners and regular gym-goers.
- Using affinity mapping and thematic clustering to identify patterns in self-tracking, progress validation, bodily discipline, and the negotiation of external influence (friends, influencers, community).

PROJECTS

Neuroanatomy SME Agent | *HuggingFace, PyTorch, LangChain* Nov. 2025

- Implemented a full embedding and retrieval pipeline — batch ingestion, multi-granularity chunk-graph construction, BioLORD embedding generation, and hybrid FAISS/Elasticsearch search with BGE reranking — powering the agent's custom LangChain-style reasoning and tool-calling.
- Developed a minimal browser UI and toolchain for chat, QA/quiz, and PDF/DOCX/PPTX export.

Multilingual SLM | *HuggingFace, PyTorch* Sept. 2025

- Built a **3B-token multilingual corpus** (English, Tamil, Mizo) from scratch with rigorous cleaning, normalization, deduplication, and deterministic train/val/test splits.
- Trained a Unigram tokenizer (96k vocab) with temperature-smoothed sampling to preserve Mizo.
- Pretrained a 140M-parameter Gemma3 (dense) LM, attaining train perplexity **2.74** at 50k steps.
- Average validation perplexity reduced from **1865** to **112** (EN: 110, TA: 187, Mizo: 38).
- Fine-tuned with LoRA on two reasoning tasks — FAQ generation: **12.7** perplexity score, and context-aware synonym substitution: **1.73** - showing excellent transfer and controlled editing behavior.

Vision Transformers | *PyTorch* Apr. 2025

- Implemented Vision Transformer (ViT) and Differential ViT on CIFAR-10 from-scratch.

- Achieved **83.69%** test accuracy after benchmarking 11 ViT Configurations, and **84.38%** test accuracy after extending to differential attention (DiffViT). Produced DINO maps and hyperparameter tuning.

Fully Convolutional Networks for Semantic Segmentation | *PyTorch, VGG* *Apr. 2025*

- Built FCN-32, 16, 8 with VGG-16 backbone and bilinear upsampling on a 13-class dataset.
- Achieved **83.3%** test mIoU on fine-tuned backbone, lesser with frozen counterparts.
- Optimized data pipelines and batch scheduling to improve training throughput and GPU utilization.

OCR CNN-RNN | *Python, NumPy, PyTorch* *Oct. 2024*

- Developed a hybrid Deep Learning pipeline for OCR tasks using CNNs for feature extraction and RNNs with integrated attention mechanisms: achieved **99.1%** ANCC (character-level accuracy).

Multi-Layer Perceptron (MLP) from Scratch | *Python, NumPy, Matplotlib* *Sept. 2024*

- Implemented MLP from scratch, that supports: customizable hidden layers, activation functions, and optimization methods, to build intuition for gradient flow and model training dynamics.

BPSK Modulation System | *MATLAB* *April 2024*

- Built a Binary Phase Shift Keying (BPSK) system for audio transmission over an AWGN and Noiseless channel, using rectangular and raised cosine pulse shaping modulation for ISI analysis.

Y86-64 ISA Processor | *Verilog* *Feb. 2024*

- Built a pipelined implementation of a Y86-64 instruction-level processor with intermediate registers to optimize instruction throughput along with hazard detection and solutions to decode the same.
- Validated functional correctness through waveform-based testbenches and timing verification.

4-bit ALU Design | *Verilog, NGSpice, MAGIC* *Nov. 2023*

- Implemented a 4-bit ALU starting with Verilog modules, progressing to gate-level simulations in NGSpice, and finally transistor-level designs (NMOS/PMOS) and layout verification using MAGIC.

ADDITIONAL EXPERIENCE

Debate Club, IIIT-H - President *Oct. 2023 - Oct. 2025*

- Grew the club to **50+ active members**, with weekly debates and theory sessions.
- Built specialized teams for logistics, marketing, and design for efficient event management to handle logistics for the increased participation; tournament training for national competitions.
- Organize flagship Debate and MUN tournaments every year, with 100+ participants.

Publication *Oct. 2025*

- Pangracious, V., Hadj-Alouane, N. B., Chu, Z., **Ramakrishnan, R.** (2025). *Impact and Perception of Generative AI and Related Automation Tools in the Gulf Region*. Palgrave Macmillan. DOI: [10.1007/978-981-96-7963-8_6](https://doi.org/10.1007/978-981-96-7963-8_6)

TEDx Talk: “Clarity in a Changing World” *March 2022*

ML Certificate (Udemy) *June 2024*

IDEOU Foundations in Design Thinking with applied user research projects *August 2021*

SKILLS

- **Languages:** Python, C, C++, R, JavaScript, HTML, CSS, L^AT_EX
- **Software Engineering:** Git, Linux/Unix Development, FastAPI, REST API, Testing & Debugging
- **ML/AI:** PyTorch, HuggingFace, Transformers, XGBoost, Model Evaluation & Optimization
- **MLOps & Cloud:** Google Cloud Platform (BigQuery, Vertex AI, Pub/Sub, Cloud Build), Kubeflow, Docker, CI/CD Pipelines, Containerized Deployment
- **Tools:** MongoDB, Pandas, NumPy, Scikit-Learn, Docker, Jupyter
- **Soft Skills:** Public speaking, Technical writing, Team leadership, Cross-functional communication